Handral Poornakanta

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Research Summary

My research is on the **experimental mechanics of slender elastic structures** — rods, ribbons, strips and beams that undergo large, geometrically nonlinear deformations. I design experiments and measurement techniques that test, and often sharpen, the mechanical models used for these structures, and I turn that understanding into devices. The work uses slender elastic structures in three ways:

- **To measure.** A geometric technique that extracts the thermal expansion coefficient from the post-buckled shape of a Joule-heated, clamped beam (*Acta Mechanica*, 2026).
- **To move.** Shape and tip control of elastic arms through load optimisation and cable (tendon) actuation, leading to a **granted patent** on a polymer 3D printer with a cable-actuated flexible arm.
- **To see.** Stereo-vision methods that reconstruct the 3D shape of strongly deformed ribbons and Möbius strips, used to test competing mechanical models against experiment.

Key Contributions to Mechanical Engineering

- **Thermal expansion via post-buckling** (Acta Mechanica, 2026; ICEM 2024 talk): a geometric route to thermal expansion coefficients of slender beams from their post-buckled shape.
- **Cable-actuated flexible-arm 3D printer** (Indian Patent 569440, granted 2025): a printer architecture in which a compliant, cable-driven arm positions the nozzle.
- **Elastica robot** (Extreme Mechanics Letters): tip control of a tendon-actuated elastic arm based on the mechanics of the elastica.
- **Elastic ribbons and Möbius strips** (JMPS 2020; Proc. R. Soc. A 2021): combined simulation, experiment and stereo-vision to evaluate mechanical models of ribbons and one-sided surfaces.
- **Shape control of the elastica** (J. Applied Mechanics): achieving target shapes of an elastic rod by optimising the applied loads.

Education

Ph.D. (Engineering), Mechanical Engineering — Aug 2022 – Present
Indian Institute of Science, Bengaluru

- Thesis: *Experimental Mechanics of Slender Elastic Structures* (Advisor: Dr. Ramsharan Rangarajan)
- Comprehensive examination passed (April 2025); research-training coursework (30 credits) completed.
- Relevant coursework: Mechanics of the Elastica (A), AI-assisted Programming (A+), Structural Optimisation, Modal Analysis, Mechanical Behaviour of Materials, Mechatronics, Robotic Actuators.

B.E., Mechanical Engineering — CGPA 8.74/10 — 2013 – 2017
B.V. Bhoomaraddi College of Engineering & Technology (VTU), Hubballi

Pre-University (12th), Science — 85% — 2013
Government P.U. College of Science & Commerce, Laxmeshwar — Dept. of Pre-University Education, Karnataka

SSLC (10th) — 82.24% — 2011
S.U.V.M. High School, Laxmeshwar — Karnataka Secondary Education Examination Board

Research Experience

Research Associate — Jun 2017 – Jul 2022
Department of Mechanical Engineering, Indian Institute of Science — Bengaluru

- Designed experiments on the inverse kinematics of the elastica, flexible robotics, and models for elastic ribbons; contributed to four journal publications.

Summer Research Fellow — Jun 2016
Indian Institute of Science — Bengaluru

- Design and study of a flexible tendon-driven robotic arm (“Elephentron”).

Journal Publications

[1] **P. Handral**, R. Rangarajan, "A geometric technique for measuring thermal expansion via post-buckling in slender beams," *Acta Mechanica* (2026). doi:10.1007/s00707-026-04798-z

[2] A. Kumar, **P. Handral**, C. S. D. Bhandari, R. Rangarajan, "More views of a one-sided surface: mechanical models and stereo vision techniques for Möbius strips," *Proceedings of the Royal Society A* 477: 20210076 (2021). doi:10.1098/rspa.2021.0076

[3] A. Kumar, **P. Handral**, C. S. D. Bhandari, A. Karmakar, R. Rangarajan, "An investigation of models for elastic ribbons: Simulations & experiments," *Journal of the Mechanics and Physics of Solids* (2020). doi:10.1016/j.jmps.2020.104070

[4] **P. Handral**, R. Rangarajan, "An elastica robot: Tip-control in tendon-actuated elastic arms," *Extreme Mechanics Letters* (2020). doi:10.1016/j.eml.2019.100584

[5] A. Nayak, **P. Handral**, R. Rangarajan, "Shape control for the elastica through load optimization," *Journal of Applied Mechanics* (2019). doi:10.1115/1.4041678

Patent

- A. Khan, **P. Handral**, C. S. D. Bhandari, R. Rangarajan, Shamrao, "Polymer 3D printer with a cable-actuated flexible arm," Indian Patent No. **569440** (Application 202341014978, filed 6 Mar 2023, **granted 31 Jul 2025**). Patentee: IISc.

Conference Presentations

- "Thermal expansion coefficients from post-buckling behavior," oral presentation, *International Conference on Experimental Mechanics (ICEM)*, 20–23 Oct 2024.
- "Shape control for the elastica through load optimization," oral presentation, *4th Indian Conference on Applied Mechanics (INCAM)*, 2019.

Honours, Awards & Fellowships

- **Dare to Dream 2.0** (DRDO) — one of the first-prize winners; facilitated by the Hon'ble Defence Minister, Shri Rajnath Singh (Oct 2021).
- 1st prize, video contest, IISc Symposium 2019.
- Admitted to the IISc Ph.D. programme through GATE (2022).

Experimental & Computational Skills

Experiments	Test-rig design, stereo vision & ArUco-marker shape measurement, CMM (PC-DMIS), DAQ, sensors & actuators, Arduino / Raspberry Pi, LabVIEW
Computation	MATLAB, Mathematica, Python, C/C++; ANSYS, Abaqus, COMSOL Multiphysics
Design	Creo, SolidWorks, AutoCAD; prototyping, 3D printing, CNC part programming
Visual design	Scientific figures & video: Adobe Creative Suite, Blender, Inkscape

References

Dr. Ramsharan Rangarajan (Ph.D. Advisor)
Department of Mechanical Engineering
Indian Institute of Science, Bengaluru

Dr. Arun Y. Patil
B.V. Bhoomaraddi College of Engg. & Tech.
(KLE Technological University), Hubballi